Invasive Lobular Carcinoma: A Multimodality Imaging Primer

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Abbreviations: IDC = invasive ductal carcinoma, ILC = invasive lobular carcinoma

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The full digital presentation is available online.

Invasive lobular carcinoma (ILC) is the second most common invasive breast cancer, accounting for 5%–15% of all invasive breast cancers. Clinical and imaging diagnosis of ILC remains challenging, and this can lead to missed or delayed diagnoses. Clinical signs of ILC may include a vague palpable area of thickening or swelling in the breast, change in nipple shape (eg, inverted nipple), dimpling of the skin, or shrinking breast. ILC is challenging to detect because of its slow-growing and insinuating nature, which results in no clinically palpable mass formation and subtle mammographic and US findings. The incidence of ILC has increased over the past 2 decades and is attributed to increased use of hormone replacement therapy in postmenopausal women and to diagnostic advances. ILC tends to occur in older women with a median age of 57 years, and it is more common in patients with lobular carcinoma in situ or atypical lobular hyperplasia.

At histologic analysis, ILC consists of small uniform cells that infiltrate circumferentially around ducts and lobules in a classic single-file pattern. Although ILC is generally of an excellent prognostic phenotype (low histologic grade, estrogen and progesterone receptor positive, human epidermal growth factor 2 [HER2] negative), it can be highly metastatic. The most common type of ILC is the classic variant, which has a slightly better prognosis than that of invasive ductal carcinoma (IDC). The pleomorphic variant has a more aggressive spread and worse prognosis.

Mammographic findings include masses, asymmetries, architectural distortion, and rarely, microcalcifications. The sensitivity of mammography is 57%–81%. In 8%–16% of cases, the mammographic findings are negative or benign. If visualized, the mammographic abnormality is frequently seen only on one view, which is more commonly the craniocaudal projection. Tumor size is often underestimated at mammography. US sensitivity ranges from 68% to 98%. Sonographic findings include masses, architectural distortion, and posterior acoustic shadowing with or without an associated mass. MRI has the highest sensitivity at 93%–95% and more accurately demonstrates the extent of disease by depicting primary tumor size and the presence of multifocal, multicentric, and bilateral disease. MRI findings include mass and nonmass enhancement. The Table summarizes the multimodality imaging findings.

TEACHING POINTS

- ILC is the second most common type of invasive breast cancer and typically manifests with subtle clinical and imaging findings, making diagnosis a challenge.
- The most common imaging manifestations of ILC include a mass with spiculated or indistinct margins at mammography and a hypoechoic irregular mass with posterior shadowing at US.
- The role of MRI in evaluation of the extent of ILC includes depicting tumor size and the presence of multifocal, multicentric, or bilateral disease.

Multimodality Imaging Findings of ILC

Modality	Imaging Characteristics and Findings
Mammography	Overall sensitivity: 57%–81% Mass with spiculated or indistinct margins is the most common appearance (44%–65%) Architectural distortion is the second most common appearance (10%–34%) Asymmetry (one view) often equal or less dense than normal breast parenchyma; classically seen on craniocaudal view only Focal asymmetry (two views) Shrinking breast appearance over time Microcalcifications uncommon (1%–25%) Round and circumscribed mass uncommon (<1%) Nipple retraction less common Normal or benign mammographic findings (8%–16%)
US	Overall sensitivity: 68%–98% Hypoechoic irregular mass with posterior shadowing is the most common appearance (60%) Frequently irregular shape and angular margins May demonstrate posterior shadowing without discrete mass Sonographically occult (10%)
MRI	Overall sensitivity: 93%–95% Solitary irregular mass with spiculated or irregular margins is the most common appearance (up to 95%) Nonmass enhancement is less frequent (69%) Kinetic features: Delayed maximum enhancement Less washout in delayed phase compared with that of IDC Can demonstrate multifocal or multicentric disease and bilateral disease not depicted with mammography or US

Compared with IDC, ILC tends to manifest with a larger size and later tumor stage, although it has a more favorable stage-matched outcome. Multifocal or multicentric and contralateral diseases are more frequently found in patients with ILC. Two-thirds of patients with ILC have metastases at the time of diagnosis. Compared with IDC, ILC is far more likely to metastasize to the peritoneum or retroperitoneum, gastrointestinal tract, urogenital tract, leptomeninges, and myocardium. The metastatic rate to the liver, bone, and pleura is comparable to that of IDC. Unusual metastatic patterns are not uncommon in ILC.

Clinical management and treatment include surgical excision with clear margins, axillary (sentinel) lymph node biopsy, and ipsilateral breast radiation. ILC is more likely to require a mastectomy than is IDC. Systemic chemotherapy is used in stage III disease or higher. However, ILC is commonly less responsive to chemotherapy than is IDC.

ILCs have an atypical clinical manifestation and imaging appearance. MRI supplements mammography to help define the extent of disease and aids presurgical planning. The online presentation reviews the mammographic, sonographic, and MRI characteristics of ILC to help radiologists accurately assess the primary tumor's size and evaluate for multicentric, multifocal, and bilateral disease, which could help prevent positive margins at breast-conserving surgery.

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Suggested Readings

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